

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1513
Additional Practice Set 1 Solutions

1. Augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array} \right) \xrightarrow[R_3-3R_1]{R_2+R_1} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array} \right) \xrightarrow{R_3-10R_2} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & 0 & -52 & -104 \end{array} \right)$$

The last matrix is in row echelon form (the form is not unique).

Now we solve the system:

$$\begin{cases} x + y + 2z = 8 \\ -y + 5z = 9 \\ -52z = -104 \end{cases}$$

Hence,

$$\begin{aligned} z &= 2 \\ y &= 5z - 9 = 1 \\ x &= 8 - y - 2z = 3 \end{aligned}$$

The system has exactly one solution: $x = 3$, $y = 1$, $z = 2$.

2. Substitute the 3 points into the equation, we get a linear system:

$$\begin{cases} d + 5c - 4b + 41a = 0 \\ d + 7c - 2b + 53a = 0 \\ d - 3c + 4b + 25a = 0 \end{cases}$$

Note that we write the variables in the order d, c, b, a so that it is easier to perform Gaussian Elimination.

Augmented matrix:

$$\begin{aligned} &\left(\begin{array}{cccc|c} 1 & 5 & -4 & 41 & 0 \\ 1 & 7 & -2 & 53 & 0 \\ 1 & -3 & 4 & 25 & 0 \end{array} \right) \xrightarrow[R_3-R_1]{R_2-R_1} \left(\begin{array}{cccc|c} 1 & 5 & -4 & 41 & 0 \\ 0 & 2 & 2 & 12 & 0 \\ 0 & -8 & 8 & -16 & 0 \end{array} \right) \\ &\xrightarrow[-\frac{1}{8}R_3]{\frac{1}{2}R_2} \left(\begin{array}{cccc|c} 1 & 5 & -4 & 41 & 0 \\ 0 & 1 & 1 & 6 & 0 \\ 0 & 1 & -1 & 2 & 0 \end{array} \right) \xrightarrow{R_3-R_2} \left(\begin{array}{cccc|c} 1 & 5 & -4 & 41 & 0 \\ 0 & 1 & 1 & 6 & 0 \\ 0 & 0 & -2 & -4 & 0 \end{array} \right) \end{aligned}$$

Now we solve the system:

$$\begin{cases} d + 5c - 4b + 41a = 0 \\ c + b + 6a = 0 \\ b + 2a = 0 \end{cases}$$

Let $a = t$ be free parameter, we get:

$$\begin{cases} a = t \\ b = -2t \\ c = -4t \\ d = -29t \end{cases}$$

$$3. \quad (a) \quad \begin{cases} 3w & - & y & = & 0 \\ 8w & & & - & 2z = 0 \\ & 2x - 2y - & z = 0. \end{cases}$$

(b) $w = \frac{1}{4}t$, $x = \frac{5}{4}t$, $y = \frac{3}{4}t$, $z = t$ where t is an arbitrary parameter.

(c) Since the number of molecules must be non-negative integers, the smallest possible t value in the general solution is $t = 4$. Hence we get the answer $w = 1$, $x = 5$, $y = 3$ and $z = 4$.

$$4. \quad (i) \quad \begin{cases} x_1 & - & x_2 & & & = & 200 \\ x_1 & & & & + & x_4 = & 500 \\ & x_2 & - & x_3 & & = & -150 \\ & & x_3 & + & x_4 & = & 450 \end{cases}$$

(ii) Since the traffic volume is non-negative, we must have:

$$x_1 = 500 - t \geq 0 \Rightarrow t \leq 500$$

$$x_2 = 300 - t \geq 0 \Rightarrow t \leq 300$$

$$x_3 = 450 - t \geq 0 \Rightarrow t \leq 450$$

$$x_4 = t \geq 0.$$

So the range of t is $0 \leq t \leq 300$.

(iii) Given that $x_1 = x_4$, we will have $500 - t = t \Rightarrow t = 250$.

So $x_1 = 250$, $x_2 = 50$, $x_3 = 200$, $x_4 = 250$.

5. (i)

$$\begin{aligned} T_1 &= \frac{1}{2}(0 + T_2) \\ T_2 &= \frac{1}{2}(T_1 + T_3) \\ T_3 &= \frac{1}{2}(T_2 + T_4) \\ T_4 &= \frac{1}{2}(T_3 + 100) \end{aligned} \Rightarrow \begin{cases} 2T_1 & - & T_2 & & & = & 0 \\ -T_1 & + & 2T_2 & - & T_3 & = & 0 \\ & & - & T_2 & + & 2T_3 & - & T_4 = 0 \\ & & & - & T_3 & + & 2T_4 & = 100 \end{cases}$$

$$(ii) \quad \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 100 \end{pmatrix}$$

$$(iii) \quad \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} 20 \\ 40 \\ 60 \\ 80 \end{pmatrix} = \begin{pmatrix} 40 - 40 \\ -20 + 80 - 60 \\ -40 + 120 - 80 \\ -60 + 160 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 100 \end{pmatrix}.$$

So $\mathbf{x}$ is a solution of the system.

(iv) No. Intuitively there can only be one possible set of temperatures at the four given nodes.

6. (i) Find the general solution of $\mathbf{B}\mathbf{x} = \mathbf{d}$ is given by:

(general solution of $\mathbf{B}\mathbf{x} = \mathbf{0}$) + (a particular solution of $\mathbf{B}\mathbf{x} = \mathbf{d}$)

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} t - s \\ s + 2t \\ 0 \\ s \\ t \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ -1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} t - s + 1 \\ s + 2t + 1 \\ -1 \\ s + 2 \\ t + 1 \end{pmatrix}$$

- (ii) The reduced row echelon form of $\mathbf{B}$ is a 3×5 matrix.

From the general solution of $\mathbf{B}\mathbf{x} = \mathbf{0}$, we can recover the three rows of the RREF as follow:

First of all, we have $x_4 = s, x_5 = t$.

$x_1 + s - t = 0 \Rightarrow x_1 + x_4 - x_5 = 0$, which give the first row of the RREF.

$x_2 - s - 2t = 0 \Rightarrow x_2 - x_4 - 2x_5 = 0$, which gives the second row of the RREF.

$x_3 = 0$ which gives the third row of the RREF.

So the RREF of $\mathbf{B}$ is

$$\begin{pmatrix} 1 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

7. (i) Let $x_2 = s$ and $x_4 = t$. We have from the REF:

$$x_5 = 0, \quad x_3 = -x_4 = -t, \quad x_1 = -x_2 - x_5 = -s.$$

The general solution is

$$\begin{array}{lcl} x_1 & = & -s \\ x_2 & = & s \\ x_3 & = & -t \\ x_4 & = & t \\ x_5 & = & 0 \end{array} \quad \text{or} \quad \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -s \\ s \\ -t \\ t \\ 0 \end{pmatrix}.$$

- (ii) General solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ is given by:

$$\begin{array}{lcl} x_1 & = & -s + 1 \\ x_2 & = & s + 2 \\ x_3 & = & -t + 3 \\ x_4 & = & t + 4 \\ x_5 & = & 5 \end{array} \quad \text{or} \quad \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -s + 1 \\ s + 2 \\ -t + 3 \\ t + 4 \\ 5 \end{pmatrix}.$$

- (iii) For $\mathbf{C}\mathbf{x} = \mathbf{0}$, yes. (Its general solution is the same as $\mathbf{A}\mathbf{x} = \mathbf{0}$.)

For $\mathbf{C}\mathbf{x} = \mathbf{b}$, no. (We do not know the REF of $(\mathbf{C} \mid \mathbf{b})$.)

$$8. \quad (a) \quad \mathbf{E}_1\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{E}_2\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 3 \end{pmatrix}, \quad \mathbf{E}_3\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & -1 & -1 \end{pmatrix}.$$

- (b) $\mathbf{E}_1$ corresponds to interchanging row 1 and row 2.

$\mathbf{E}_2$ corresponds to multiplying row 3 by constant 3.

$\mathbf{E}_3$ corresponds to adding -2 times of row 1 to row 3.

9. (a) Yes. Let $\mathbf{M}$ be the matrix $\mathbf{A} + \mathbf{A}^T$. Take the transpose of $\mathbf{M}$:

$$\mathbf{M}^T = (\mathbf{A} + \mathbf{A}^T)^T = \mathbf{A}^T + (\mathbf{A}^T)^T = \mathbf{A}^T + \mathbf{A} = \mathbf{A} + \mathbf{A}^T = \mathbf{M}$$

Since $\mathbf{M}^T = \mathbf{M}$, this implies $\mathbf{M} = \mathbf{A} + \mathbf{A}^T$ is symmetric.

- (b) Yes. Let $\mathbf{N}$ be the matrix $\mathbf{A}\mathbf{A}^T$. Take the transpose of $\mathbf{N}$:

$$\mathbf{N}^T = (\mathbf{A}\mathbf{A}^T)^T = (\mathbf{A}^T)^T \mathbf{A}^T = \mathbf{A}\mathbf{A}^T = \mathbf{N}$$

Again, $\mathbf{N}^T = \mathbf{N}$. This implies $\mathbf{N} = \mathbf{A}\mathbf{A}^T$ is symmetric.

10. (i)

$$\begin{aligned} j_{n+1} &= 1.6a_n \\ a_{n+1} &= 0.3j_n + 0.8a_n \end{aligned} \implies \begin{pmatrix} j_{n+1} \\ a_{n+1} \end{pmatrix} = \begin{pmatrix} 0 & 1.6 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} j_n \\ a_n \end{pmatrix}$$

$$(ii) \mathbf{A}^{-1} = \frac{1}{-0.48} \begin{pmatrix} 0.8 & -1.6 \\ -0.3 & 0 \end{pmatrix}$$

$$(iii) \mathbf{x}_{2019} = \begin{pmatrix} 480 \\ 360 \end{pmatrix}.$$

$$\mathbf{x}_{2020} = \begin{pmatrix} 0 & 1.6 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} 480 \\ 360 \end{pmatrix} = \begin{pmatrix} 576 \\ 432 \end{pmatrix}.$$

$$\text{So } j_{2020} = 576, \quad a_{2020} = 432.$$

$$\mathbf{x}_{2018} = \frac{1}{-0.48} \begin{pmatrix} 0.8 & -1.6 \\ -0.3 & 0 \end{pmatrix} \begin{pmatrix} 480 \\ 360 \end{pmatrix} = \begin{pmatrix} 400 \\ 300 \end{pmatrix}.$$

$$\text{So } j_{2018} = 400, \quad a_{2018} = 300.$$